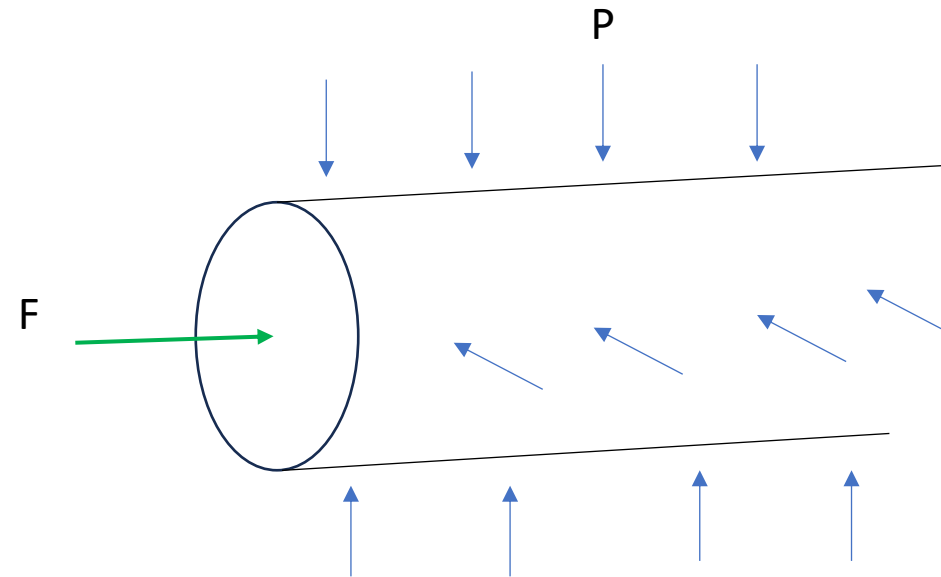


# Buckling Method for Composite Cylinders Under Axial Loading and External Pressure

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$$N_z = -\frac{pR}{2}$$

$$N_\theta = -p \cdot R$$



## Ritz Method for **Stability Analysis** of Composite Cylinders

- Potential of the external load

$$V = -\frac{1}{2} \int_S N_z \left( \frac{\partial w}{\partial z} \right)^2 + N_\theta \frac{1}{R^2} \left( \frac{\partial w}{\partial \theta} \right)^2 dS$$



$$\frac{\partial w}{\partial z} = [N_{w,z}] \{q_w\}$$

$$\frac{\partial w}{\partial \theta} = [N_{w,\theta}] \{q_w\}$$

$$V = -\frac{1}{2} \int_S N_z \begin{Bmatrix} \{q_u\} \\ \{q_v\} \\ \{q_w\} \\ \{q_{\beta z}\} \\ \{q_{\beta \theta}\} \end{Bmatrix}^T \begin{bmatrix} [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [N_{w,z}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \end{bmatrix}^T \begin{Bmatrix} \{q_u\} \\ \{q_v\} \\ \{q_w\} \\ \{q_{\beta z}\} \\ \{q_{\beta \theta}\} \end{Bmatrix} + N_\theta \frac{1}{R^2} \begin{Bmatrix} \{q_u\} \\ \{q_v\} \\ \{q_w\} \\ \{q_{\beta z}\} \\ \{q_{\beta \theta}\} \end{Bmatrix}^T \begin{bmatrix} [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [N_{w,\theta}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \end{bmatrix}^T \begin{Bmatrix} \{q_u\} \\ \{q_v\} \\ \{q_w\} \\ \{q_{\beta z}\} \\ \{q_{\beta \theta}\} \end{Bmatrix} dS$$

$$W = -V = \frac{1}{2} \int_S N_z \{q\}^T [H_z]^T [H_z] \{q\} + N_\theta \frac{1}{R^2} \{q\}^T [H_\theta]^T [H_\theta] \{q\} dS \quad W = \frac{1}{2} \int_S \{q\}^T \left\{ N_z [H_z]^T [H_z] + N_\theta \frac{1}{R^2} [H_\theta]^T [H_\theta] \right\} \{q\} dS$$

$$[H_z] = \begin{bmatrix} [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [N_{w,z}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \end{bmatrix}$$

$$[H_\theta] = \begin{bmatrix} [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [N_{w,\theta}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \end{bmatrix}$$

$$N_z = -\frac{p \cdot R}{2} \quad *$$

$$N_\theta = -p \cdot R$$

$$N_z = \frac{N_\theta}{2} = \lambda$$

$$W = \frac{1}{2} \int_S \{q\}^T \left\{ N_z [H_z]^T [H_z] + N_\theta \frac{1}{R^2} [H_\theta]^T [H_\theta] \right\} \{q\} dS$$

$$W = \frac{\lambda}{2} \int_S \{q\}^T \left\{ [H_z]^T [H_z] + 2 \frac{1}{R^2} [H_\theta]^T [H_\theta] \right\} \{q\} dS$$

\*

$$\sigma_z = \frac{F}{A} = \frac{P \pi R^2}{2 \pi R t} = \frac{N_z}{t}$$

$$N_z = -\frac{p \cdot R}{2}$$

- In the equilibrium

$$\delta\pi_p = \delta U - \delta W = 0$$

$$\delta W = \lambda \int_S \{\delta q\}^T \left\{ [H_z]^T [H_z] + 2 \frac{1}{R^2} [H_\theta]^T [H_\theta] \right\} \{q\} dS$$

$$\delta\pi = \{\delta q\}^T \left\{ \int_S [H]^T [C^*] [H] dS + \lambda \int_S \{q\}^T \left\{ [H_z]^T [H_z] + 2 \frac{1}{R^2} [H_\theta]^T [H_\theta] \right\} \{q\} dS \right\} \{q\} = 0$$

$$[K_{Gz}] = \int_S \left\{ [H_z]^T [H_z] + \frac{2}{R^2} [H_\theta]^T [H_\theta] \right\} dS$$

$$\{[K] + \lambda[K_{Gz}]\} \{q\} = \{0\}$$

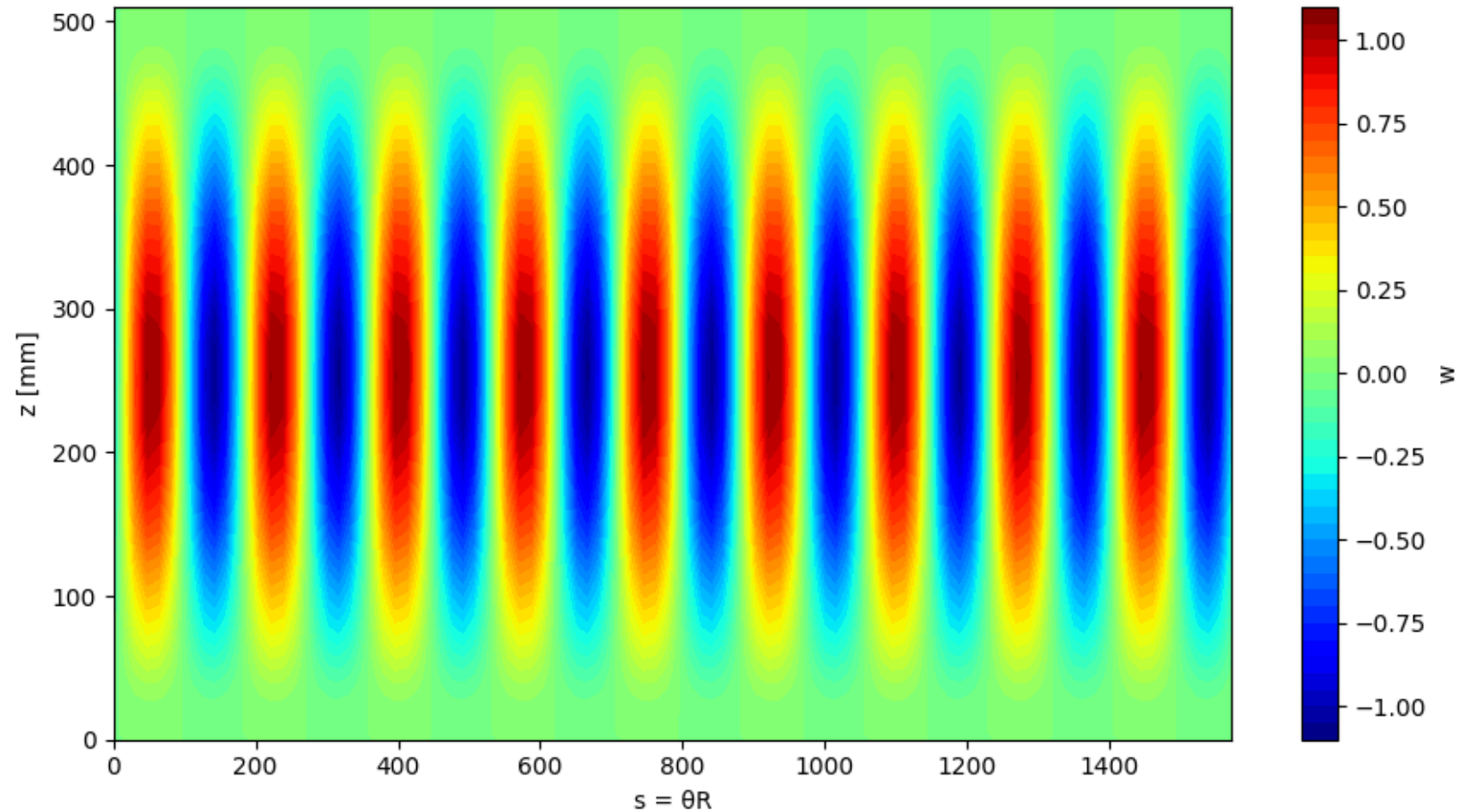
$$N_z = -\frac{p \cdot R}{2} = \lambda$$

$$p_{cr} = -2 \frac{\lambda}{R}$$

# Results



$P_{cr}$  [kN]: -10.2379  
 $pressure_{cr}$  [MPa]: -0.0521

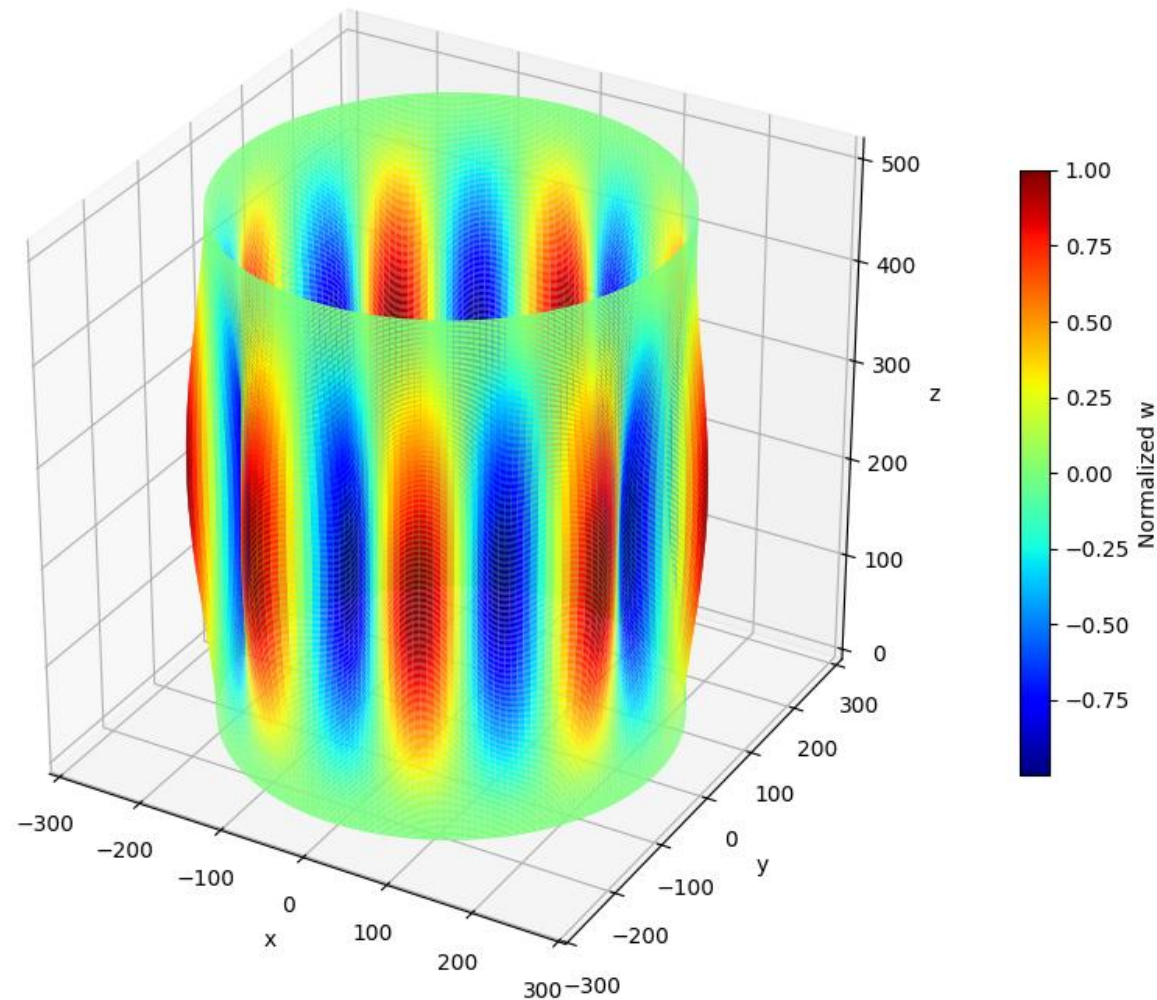


# Results



$P_{cr}$  [kN]: -10.2379  
 $pressure_{cr}$  [MPa]: -0.0521

Cylinder buckling mode shape



P. R. Santos and M. V. Donadon, “A semi-analytical model for buckling and stress analyses of pressurized composite cylinders,” *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, vol. 45, no. 9, Sep. 2023, doi: 10.1007/s40430-023-04350-3.

P. R. dos Santos and M. V. Donadon, “A semi-analytical model for buckling and stress analyses of pressurized composite cylinders,” *Proceedings of the 6th Brazilian Conference on Composite Materials*, pp. 65–70, 2022, doi: 10.29327/566492

P. R. Santos, M. V. Donadon, and M. A. Arbelo, “A Semi-Analytical Model for Stability Analyses of Composite Cylinders Accounting for Loading and Geometrical Imperfections Effects,” in *Proceedings of the 7th Brazilian Conference on Composite Materials (BCCM)*, Universidade de Brasília, 2024, pp. 324–331